The Poisson Model in Football Betting

Godfrey Tshehla

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Abstract

This paper provides a brief introduction to the use of the Poisson distribution in football betting. Expected scoring rates are estimated from team attacking and defensive strengths and used to construct probabilities for match scorelines. These probabilities can subsequently be converted into fair betting odds and compared with bookmaker prices.

Keywords: Poisson distribution, football, betting, expected goals, correct score, fair odds.

1 Introduction

Statistical models provide a systematic method for estimating the probabilities of football outcomes. One of the most established approaches is the Poisson model. Maher [1982] showed that football scores can be modelled using scoring rates determined by the attacking and defensive abilities of the competing teams.

The model is particularly useful for betting because it produces a probability distribution over possible scores rather than only predicting a winner. The resulting distribution can be used to price correct-score, match-result, total-goals, and related markets.

2 Poisson Goal Model

Let X denote the number of goals scored by a team. Under the Poisson assumption,

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \tag{2.1}$$

where k is the number of goals and λ is the expected number of goals scored.

For a home team, a simple expected-goals specification is

$$\lambda_H = L_H A_H D_A, \tag{2.2}$$

where L_H is the league average number of home goals, A_H is the home team's attacking strength, and D_A represents the away team's defensive strength.

Similarly,

$$\lambda_A = L_A A_A D_H. \tag{2.3}$$

Attack strength can be measured relative to the league average. For example,

$$A_H = \frac{\text{Team home goals per match}}{\text{League home goals per match}}. \tag{2.4}$$

A value above one therefore represents an attack scoring above the league average.

3 Score Probabilities

In the basic model, home and away goals are treated as independent Poisson random variables. The probability of a particular final score (x, y) is therefore

$$P(H = x, A = y) = P(H = x)P(A = y). \tag{3.1}$$

Calculating this probability over a range of home and away goals creates a score matrix. The matrix can be used directly for correct-score betting or aggregated to calculate probabilities for home wins, draws, away wins, both teams to score, and over/under goal markets.

4 Betting Application

Model probabilities can be converted into fair decimal odds using

$$O_{\text{fair}} = \frac{1}{P_{\text{model}}}. \tag{4.1}$$

A model probability of 0.10, for example, corresponds to fair odds of 10.00.

The model price can then be compared with the bookmaker price. Expected value is

$$EV = P_{\text{model}} O_{\text{book}} - 1. \tag{4.2}$$

Positive expected value occurs when the bookmaker offers sufficiently higher odds than those implied by the model. Consequently, the objective is not simply to identify the most likely result but to identify outcomes whose available price exceeds their estimated fair value.

5 Limitations

The basic Poisson model assumes conditional independence between the goals scored by the two teams. Football matches may violate this assumption because scoring behaviour changes with match state.

The model is also highly dependent on the quality of the estimates of λ_H and λ_A. Historical goals can

be noisy indicators of underlying team strength, while injuries, squad changes, opponent quality, and recent performance can affect future scoring rates.

Dixon and Coles [1997] extended the Poisson framework by introducing a correction for dependence in low-scoring football results and allowing team strengths to vary through time.

6 Conclusion

The Poisson distribution provides a simple and interpretable framework for football betting. Expected scoring rates are transformed into a complete distribution of possible match scores, from which probabilities and fair odds for several betting markets can be obtained.

Its simplicity makes the Poisson model a useful baseline. More advanced approaches, including the Dixon–Coles and bivariate Poisson models, can subsequently be compared with this baseline using out-of-sample testing.

References

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